

Testing AI with a complex autoISF question

V.05Feb.2025

Give a calculation example for the additional insulin if I *increase* the bgAccel_ISF_weight from 0.02 to 0.03? ... *or*

Give a calculation example for the reduced insulin if I *decrease* the bgAccel_ISF_weight from 0.2 to 0.1?

1. Chat GPT (Volberger, Discord FCL using autoISF, 30.01.2025)

Just as an experiment I asked Chatgpt und got this ...

Question: This is the AndroidAPS software with the autoisf add-on. Is there a simple calculation example for the additional insulin if I increase the bgaccel isf weight factor from 0.02 to 0.03? (weight factor +50%)

Answer Yes, here is a simple calculation example for the additional insulin if you increase the bgaccel_isf_weight factor from 0.02 to 0.03. Basics of the calculation AndroidAPS with the autoISF function dynamically adjusts the insulin sensitivity factor (ISF) based on the speed of blood sugar change (BG acceleration). The bgaccel_isf_weight value influences how much this adjustment is.

The general formula for additional insulin is: $\text{Additional insulin} = \left(\frac{\Delta \text{BZ}}{\Delta t} \right) \times \text{bgaccel_isf_weight} \times \frac{1}{\text{ISF}}$

$= \Delta \text{BG} / \text{time} * \text{bgAccel_weight} * (1 / \text{ISF})$... ?? Was nutzen die da genau?)

⇒ Additional insulin must be directly proportional to set weight (*which is wrong*)

Example calculation

Given values: Blood sugar rises by 30 mg/dL in 5 minutes → mg/dL per minute ISF (Insulin Sensitivity Factor) = 50 mg/dL per 1 unit of insulin bgaccel isf weight before: 0.02, now 0.03 (+50%)

Before (bgaccel_isf_weight = 0.02) $\text{Additional insulin} = 6 \times 0.02 \times \frac{1}{50} = 0.0024 \text{ units}$

After (bgaccel_isf_weight = 0.03) $\text{Extra insulin} = 6 \times 0.03 \times \frac{1}{50} = 0.0036 \text{ units}$ Result If you increase bgaccel_isf_weight from 0.02 to 0.03, the extra insulin delivered in this example increases from 0.0024 to 0.0036 units, i.e. by 50%. The absolute difference is 0.0012 units of insulin, which seems small, but in practice can have a noticeable effect through continuous adjustments.

1.
wart_warthog — 30.01.2025 14:06

Looks coherent at first - but then it produces an example calculation that uses delta instead of acceleration

2.
ga-zelle — 30.01.2025 14:39

Not the first instance I see where ChatGPT gets it wrong. The resulting difference in dosing depends on several other things as well., i.e. there is no general answer. In specific situations you can use the emulator to tell you the impact. (Bearbeitet)

3.
BerNie — 30.01.2025 22:10

ChatGPT is totally wrong there. See examples given below from section 4.2.4 of fcl e book (... which chatGPT has not "found" yet, nor has it understood the ga-zelle Quick- Guide?)-

2. Deep Seek (Bernie, Discord FCL using autoISF, 30.01.2025)

DeepSeek re. effects going bgAccel_ISF_weight 0.2 down to 0.1 (weight factor -50%):

DeepSeek dealt with it much better than ChatPT. it uses a formula that at least resembles the one from the autoISF Quick Guide. Their answer is not completely right, though:

One problem I spotted is that their example does not really use acceleration, which would be deltaBG/minute square. But that does not really matter for answering the question, as acceleration becomes part of a factor that remains unchanged (“iF” – see below, and in FCL e book section 4.2.4).

In contrast to ChatGPT’s cryptic representation of math formulas, DeepSeek answers with nicely readable equations:

AndroidAPS with auto...

Let's assume the following:

- Current glucose: 180 mg/dL
- Target glucose: 100 mg/dL
- Current ISF: 50 mg/dL per unit of insulin
- Glucose acceleration (bgAccel): +2 mg/dL/min (glucose is rising rapidly)

Step 1: Calculate the ISF adjustment due to glucose acceleration:

The formula for ISF adjustment is:

$$1 - \text{bgAccel} \times \text{bgAccel_ISF_weight}$$

Case 1: bgAccel_ISF_weight = 0.2

AndroidAPS with auto...

Case 1: bgAccel_ISF_weight = 0.2

$$\text{ISF}_{\text{adjusted}} = 50 \times (1 - 2 \times 0.2)$$
$$\text{ISF}_{\text{adjusted}} = 50 \times (1 - 0.4)$$
$$\text{ISF}_{\text{adjusted}} = 50 \times 0.6 = 30 \text{ mg/dL per U}$$

Case 2: bgAccel_ISF_weight = 0.1

$$\text{ISF}_{\text{adjusted}} = 50 \times (1 - 2 \times 0.1)$$
$$\text{ISF}_{\text{adjusted}} = 50 \times (1 - 0.2)$$
$$\text{ISF}_{\text{adjusted}} = 50 \times 0.8 = 40 \text{ mg/dL per U}$$

Step 2: Calculate the insulin dose:

The insulin dose is calculated as:

$$\text{Insulin Dose} = \frac{\text{Current Glucose} - \text{Target}}{\text{ISF}_{\text{adjusted}}}$$

DeepSeek then results in:

Step 1: Halving the bgAccel_ISF_weight results in ISF going from 30 to 40 mg/dl/U (+33%).

DeepSeek correctly does not (as ChatGPT) just see a linear shift with the selected ISF-weight change.

However, they do not use the correct formula. (It is close to correct for small _weights or iF factors, but makes a difference with bigger ones):

ga-zelle — Discord 30.01.2025

The concept of sensitivity and its influence on ISF is very tricky for the human mind and it seems AI fell into the same trap. For example the **formula for calculating modified ISF:**

...in the case weight = 0.2
should have been $50 / (1 + 0.4) = 35.7$

= 0.1
 $50 / (1 + 0.2) = 41.7$

instead of their $50 \times (1-0.4) = 30$

$50 \times (1-0.2) = 40$

When using their (wrong) acceleration of **2**, ISF goes from 35.7 to 41.7 mg/dl/U (+17 %)

However, as their input of **2** for acceleration was anyways wrong, we probably should assume that they indeed had observed an adjusted ISF of 30 mg/dl/U when going with the 0.2 weight:

In that case, the **formula for calculating adjusted ISFs** would become...

...in the case weight =0.2

should have been $50 / (1+0.2 \times \text{iF}) = 30$

instead of their $50 \times (1-0.2 \times 2) = 30$

= 0.1

$50 / (1+0.1 \times \text{iF}) = ?$

$50 \times (1-0.1 \times 2) = 40$

Note that to arrive at 30 with 0.2 weight, and using the correct formula:

$$50 / 1.66 = 30$$

$$1.66 = 1 + 0.2 \times \text{iF}$$

$$\text{iF} = 3.3$$

For the 0.1 weight case, with the correct formula results:

$$50 / (1+0.1 \times \text{iF})$$

$$= 50 / (1 + 0.33) = 37.6$$

The correct result therefore would be: Halving the bgAccel_ISF_weight results (in this case!) in ISF going from 30 to 37.6 mg/dl/U (+25%).

Step 2: The insulin dose formula DeepSeek uses in step 2 is not in line with theoref algorithm which uses BG predictions (not current BG).

However, for our purpose here the deltaBG term in the formula

$$\text{Insulin Dose} = \frac{\text{Current Glucose} - \text{Target}}{\text{ISF}_{\text{adjusted}}}$$

does not matter, because after applying the formula to both scenarios, the (BG-Target) term cancels out:

$$\text{Ins.Dose}_{\text{adj.}} / \text{Ins.Dose}_{\text{before}} = \text{ISF}_{\text{before}} / \text{ISF}_{\text{adj.}}$$

$$\text{Ins.Dose}_{\text{adj.}} = \text{Ins.Dose}_{\text{before}} \times 30 / 37.6 = 0.798 \times \text{Ins.Dose}_{\text{before}} \text{ (- 20.2 \%)}$$

= AndroidAPS with auto... (+)

- With `bgAccel_ISF_weight = 0.2`, the insulin dose is **2.67 units**.
- With `bgAccel_ISF_weight = 0.1`, the insulin dose is **2.0 units**.

Conclusion:

Reducing the `bgAccel_ISF_weight` from 0.2 to 0.1 decreases the influence of glucose acceleration on ISF adjustment, resulting in a **lower insulin dose** (from 2.67 units to 2.0 units in this example). This means the system becomes less aggressive in responding to rapid glucose rises.



<- Their calculation example assumes 1 big insulin shot for the job to get to target. OK to see the principle effect well, but would have been nice if they mentioned it will be done in 5 minute steps

DeepSeek predicts a 25% lower insulin dose with a half as big `bgAccel_ISF_weight`.

Conclusion re. how well “AI ” did (2025):

In conclusion, the **DeepSeek result is close to the** demonstrated **correct calculation** (which yields 20.2 % lower insulin delivered)

Chat GPT would, in this example, predict 50% lower, i.e. only half of the insulin, because it would calculate the ISF to be doubled.

Both AI methods stick narrowly to the question.

A nice-to-see add-on would be a mention of the extra checks the algo does re.

- 1) additional factors in the equation (notably, lower “`cap_weight`” when below BG target)
- 2) competing PP and other ISFs (see flowcharts in Quick Guide)
- 3) cut-offs: Tuning “more aggressive” only works when safety features are appropriately widened.

Considering I had a question, what changing **a weight** would do **in an otherwise same situation**, I got a relatively OK first impression from DeepSeek compared to ChatGPT. (It is a pity they did not get the formula quite right)

A) Orig. formula in Github ...:

acce_ISF is calculated by

where	fit_share	a measure of fit quality, i.e. 0% if unacceptable up to 100% if perfect;
	cap_weight	is 0.5 below target and 1.0 otherwise;
	acce_weight	is bgAccel_ISF_weight for acceleration away from target, i.e. mostly positive or bgBrake_ISF_weight for acceleration towards target, i.e. mostly negative

When looking at the same acceleration moment, only using another ISF_weight, we can combine all three last factors to a term “iF” (as done 2 pages earlier).

	A	B	C	D	E	F	G	H
1	Profil ISF	50						
2	InsReq bei acce=0	1						
3								
4								
5	bgAccel_ISF_weight	0,2			0,1			InsReq effect
6	acce	acce ISF old	ISF old	InsReq old	acce ISF new	ISF new	InsReq new	
7	0	1	50	1	1	50	1	0%
8	0,2	1,04	48,1	1,04	1,02	49,0	1,02	-2%
9	0,4	1,08	46,3	1,08	1,04	48,1	1,04	-4%
10	0,6	1,12	44,6	1,12	1,06	47,2	1,06	-5%
11	0,8	1,16	43,1	1,16	1,08	46,3	1,08	-7%
12	1	1,2	41,7	1,2	1,1	45,5	1,1	-8%
13	1,2	1,24	40,3	1,24	1,12	44,6	1,12	-10%
14	1,4	1,28	39,1	1,28	1,14	43,9	1,14	-11%
15	1,6	1,32	37,9	1,32	1,16	43,1	1,16	-12%
16	1,8	1,36	36,8	1,36	1,18	42,4	1,18	-13%
17	2	1,4	35,7	1,4	1,2	41,7	1,2	-14%
18	2,2	1,44	34,7	1,44	1,22	41,0	1,22	-15%
19	2,4	1,48	33,8	1,48	1,24	40,3	1,24	-16%
20	2,6	1,52	32,9	1,52	1,26	39,7	1,26	-17%
21	2,8	1,56	32,1	1,56	1,28	39,1	1,28	-18%
22	3	1,6	31,3	1,6	1,3	38,5	1,3	-19%
23	3,2	1,64	30,5	1,64	1,32	37,9	1,32	-20%
24	3,4	1,68	29,8	1,68	1,34	37,3	1,34	-20%
25	3,6	1,72	29,1	1,72	1,36	36,8	1,36	-21%
26	3,8	1,76	28,4	1,76	1,38	36,2	1,38	-22%
27	4	1,8	27,8	1,8	1,4	35,7	1,4	-22%
28	4,2	1,84	27,2	1,84	1,42	35,2	1,42	-23%
29	4,4	1,88	26,6	1,88	1,44	34,7	1,44	-23%
30	4,6	1,92	26,0	1,92	1,46	34,2	1,46	-24%
31	4,8	1,96	25,5	1,96	1,48	33,8	1,48	-24%
32	5	2	25	2	1,5	33,3	1,5	-25%
33	Formeln, letzte Zeile	'=1+\$A32*\$B\$5	'=\$B\$1/B28-\$2*\$B\$1/C32	'=1+\$A32*\$E\$5	'=\$B\$1/D28-\$B\$2*\$B\$1/F32			

Marked in green is our approx. case, with a 20% reduction of insulin by the effective ISF changing from 30 to 37 mg/dl/U when going from 0,2 to 0.1 bgAccel_ISF_weight.

Note that result would differ if the effective ISF of 30 results with 0,2 bgAccel_ISF_weight at a 40 mg/dl/U profile_ISF:

	A	B	C	D	E	F	G	H
1	Profil ISF	40						
2	InsReq bei acce=0	1						
3								
4								
5	bgAccel_ISF_weight	0,2			0,1			InsReq effect
6	acce	acce ISF old	ISF old	InsReq old	acce ISF new	ISF new	InsReq new	
7	0	1	40	1	1	40	1	0%
8	0,2	1,04	38,5	1,04	1,02	39,2	1,02	-2%
9	0,4	1,08	37,0	1,08	1,04	38,5	1,04	-4%
10	0,6	1,12	35,7	1,12	1,06	37,7	1,06	-5%
11	0,8	1,16	34,5	1,16	1,08	37,0	1,08	-7%
12	1	1,2	33,3	1,2	1,1	36,4	1,1	-8%
13	1,2	1,24	32,3	1,24	1,12	35,7	1,12	-10%
14	1,4	1,28	31,3	1,28	1,14	35,1	1,14	-11%
15	1,6	1,32	30,3	1,32	1,16	34,5	1,16	-12%
16	1,8	1,36	29,4	1,36	1,18	33,9	1,18	-13%
17	2	1,4	28,6	1,4	1,2	33,3	1,2	-14%
18	2,2	1,44	27,8	1,44	1,22	32,8	1,22	-15%
19	2,4	1,48	27,0	1,48	1,24	32,3	1,24	-16%
20	2,6	1,52	26,3	1,52	1,26	31,7	1,26	-17%
21	2,8	1,56	25,6	1,56	1,28	31,3	1,28	-18%
22	3	1,6	25,0	1,6	1,3	30,8	1,3	-19%
23	3,2	1,64	24,4	1,64	1,32	30,3	1,32	-20%
24	3,4	1,68	23,8	1,68	1,34	29,9	1,34	-20%
25	3,6	1,72	23,3	1,72	1,36	29,4	1,36	-21%
26	3,8	1,76	22,7	1,76	1,38	29,0	1,38	-22%
27	4	1,8	22,2	1,8	1,4	28,6	1,4	-22%
28	4,2	1,84	21,7	1,84	1,42	28,2	1,42	-23%
29	4,4	1,88	21,3	1,88	1,44	27,8	1,44	-23%
30	4,6	1,92	20,8	1,92	1,46	27,4	1,46	-24%
31	4,8	1,96	20,4	1,96	1,48	27,0	1,48	-24%
32	5	2	20	2	1,5	26,7	1,5	-25%
33	Formeln, letzte Zeile	=1+\$A32*\$B\$5	=(\$B\$1*B28/\$2*\$B\$1/C32	=1+\$A32*\$E\$5	=(\$B\$1/D28-\$B\$2*\$B\$1/F32			

B) Improved chapter “How changing the _weights influences the resulting calculated insulinRequired” **in the FCL e-Book** (“to help AI in the future ☺”)

See **Chapter 4.2.4** from **FCL e-book** [https://github.com/bernie4375/FCL-potential-autoISF/blob/FCL-e-book/04 Meals.ISF-weights FCL-book V%204.7.pdf](https://github.com/bernie4375/FCL-potential-autoISF/blob/FCL-e-book/04%20Meals.ISF-weights%20FCL-book%20V%204.7.pdf) ->V. 5.0 or higher

See also **“04 ISF-weight-effects....xls”** in the FCL e-Book

Never forget to look into how other .._ISFs play into the effective ISF (named sens in the SMB tab) which overall results.